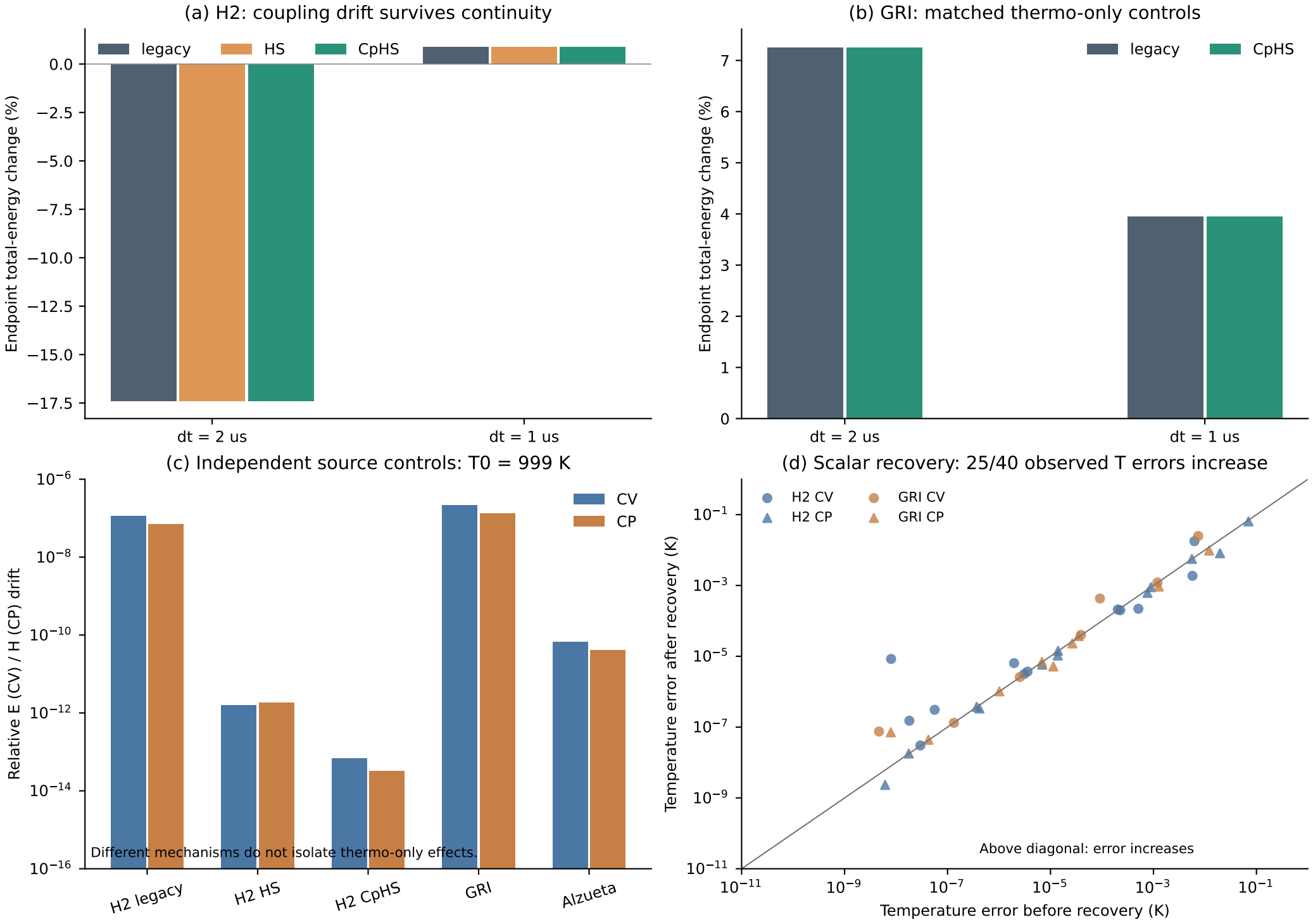
(a) H2: coupling drift survives continuity
legacy
HS
CpHS
Endpoint total-energy change (%)
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
dt = 2 us
dt = 1 us
(b) GRI: matched thermo-only controls
legacy
CpHS
Endpoint total-energy change (%)
7
6
5
4
3
2
1
0
dt = 2 us
dt = 1 us
(c) Independent source controls: T0 = 999 K
CV
CP
Relative E (CV) / H (CP) drift
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
Different mechanisms do not isolate thermo-only effects.
H2 legacy
H2 HS
H2 CpHS
GRI
Alzueta
(d) Scalar recovery: 25/40 observed T errors increase
H2 CV
H2 CP
GRI CV
GRI CP
Temperature error after recovery (K)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
Above diagonal: error increases
10^{-11}
10^{-9}
10^{-7}
10^{-5}
10^{-3}
10^{-1}
Temperature error before recovery (K)